

Analog Avengers

Data Analysis

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DATASET DESCRIPTION

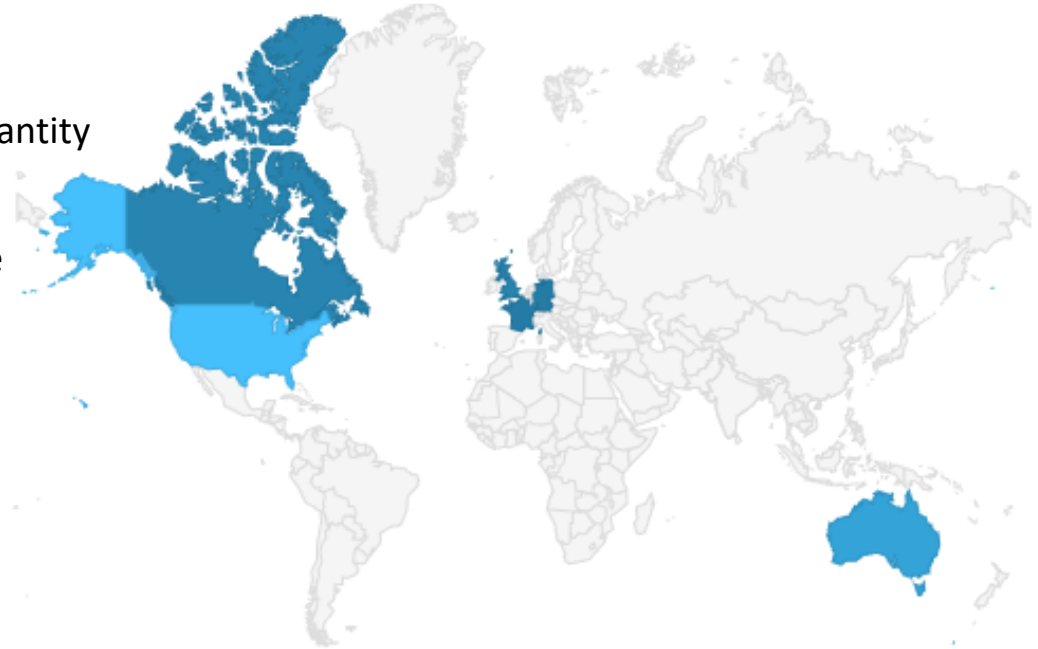
Bike sales in Europe

- From Kaggle
- > 100k entries
- 18 columns
- Ranging from 2011 to 2016
- 6 different countries



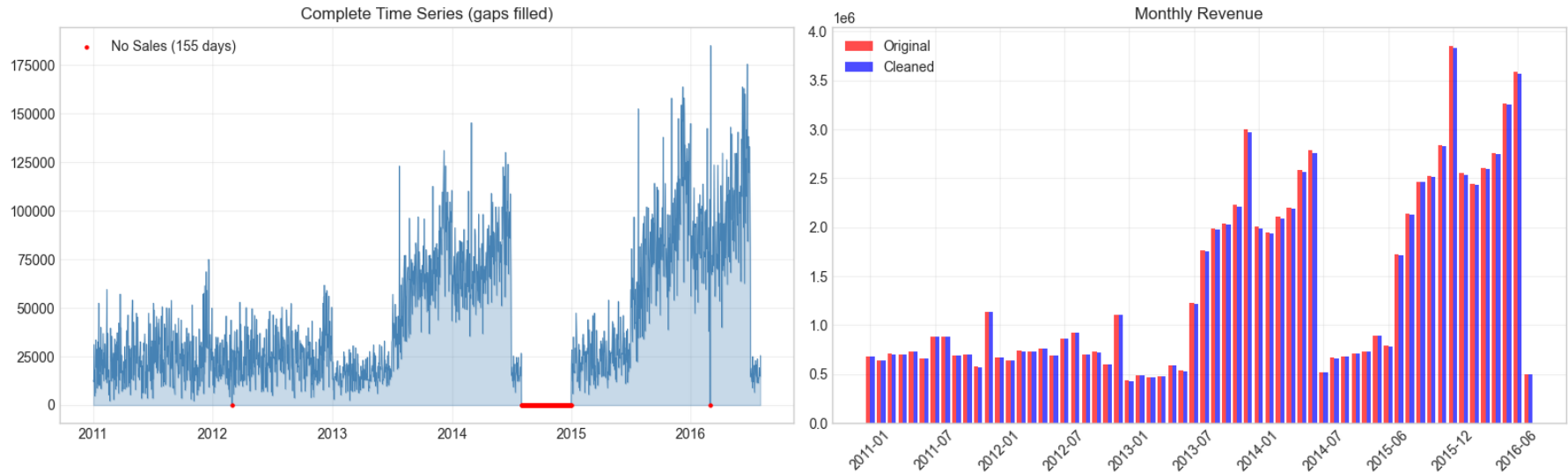
Basic Analysis

- Date
- Day
- Month
- Year
- Customer_Age
- Age_Group
- Customer_Gender
- Country
- State
- Product_Category
- Sub_Category
- Product
- Order_Quantity
- Unit_Cost
- Unit_Price
- Profit
- Cost
- Revenue



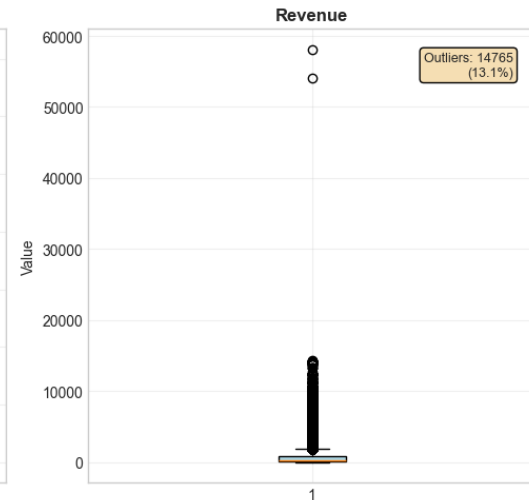
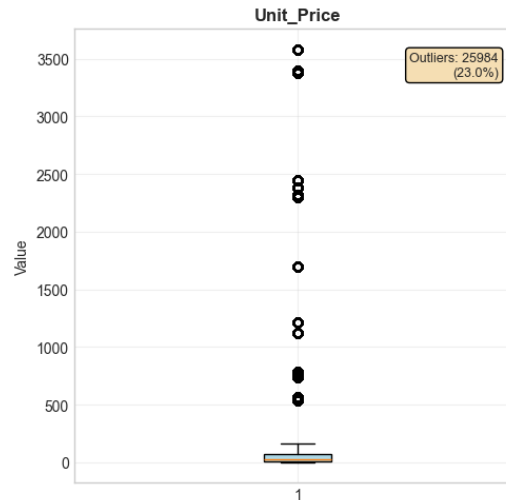
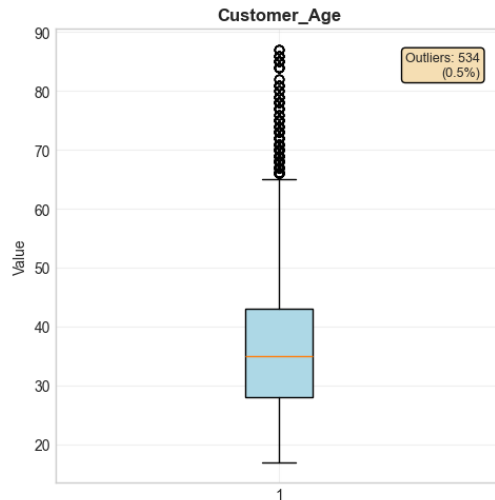
Data Quality

- Missing entries, but no missing data



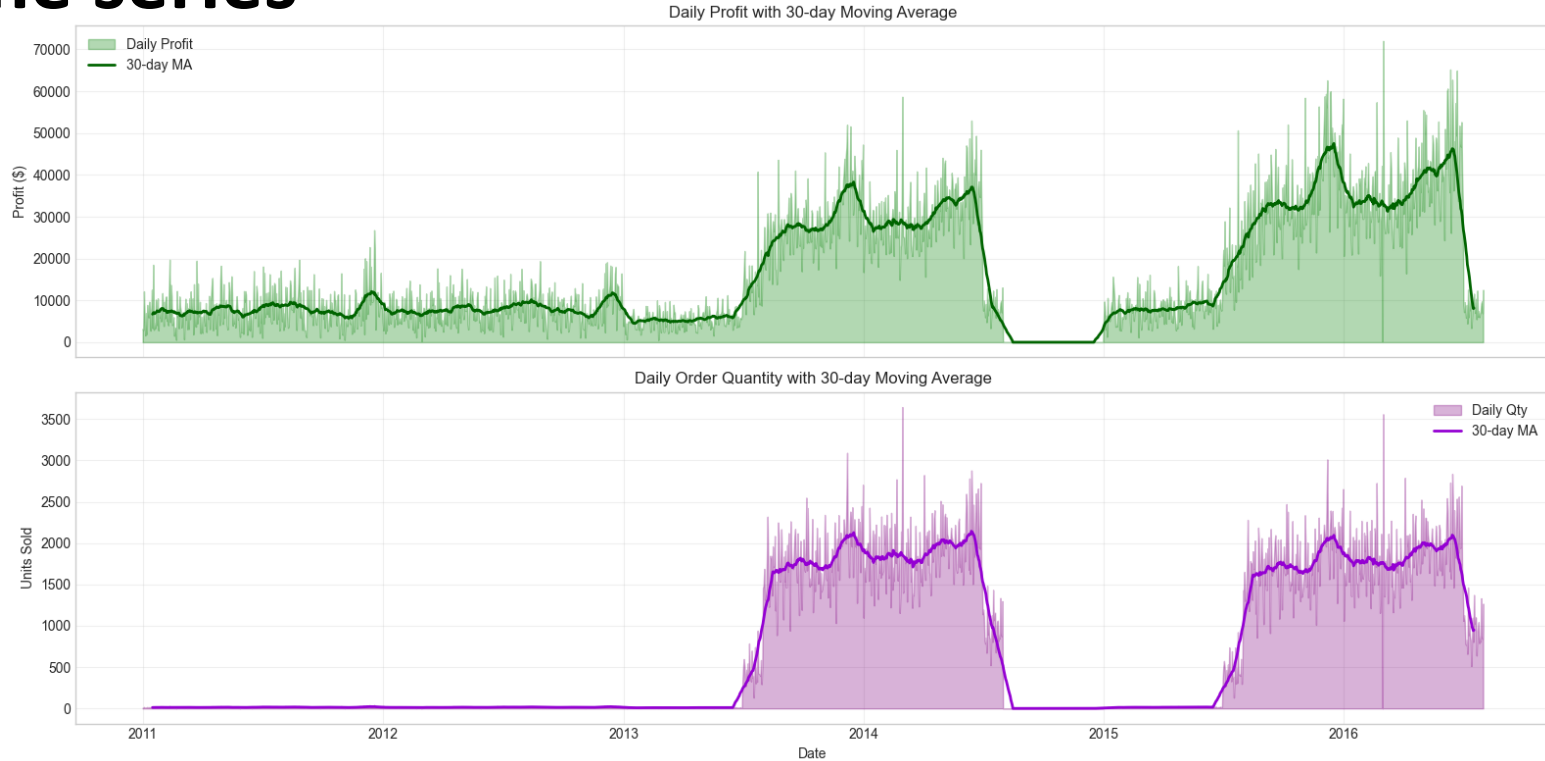
Data Quality

- Outliers detected, but not removed

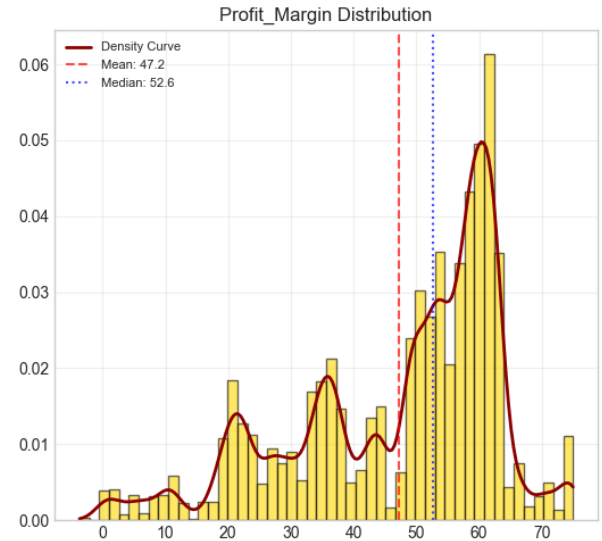
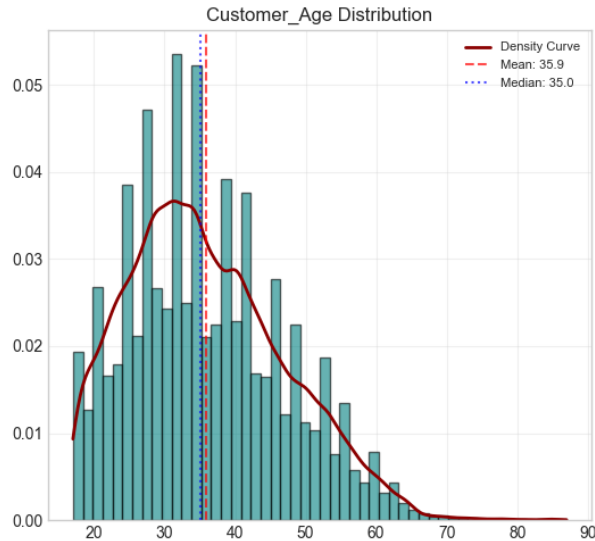
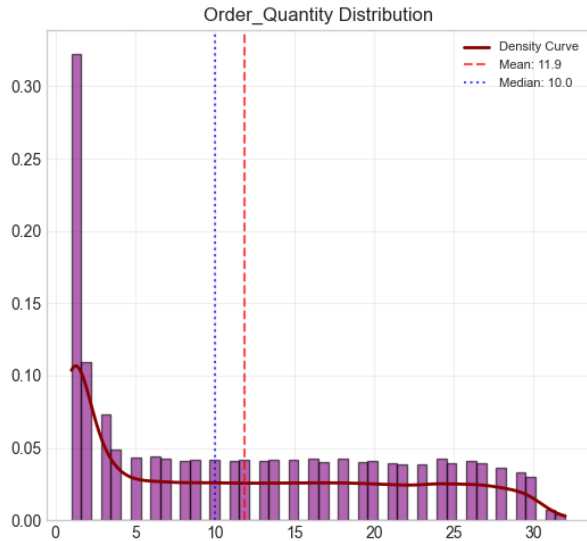


VISUALIZATION

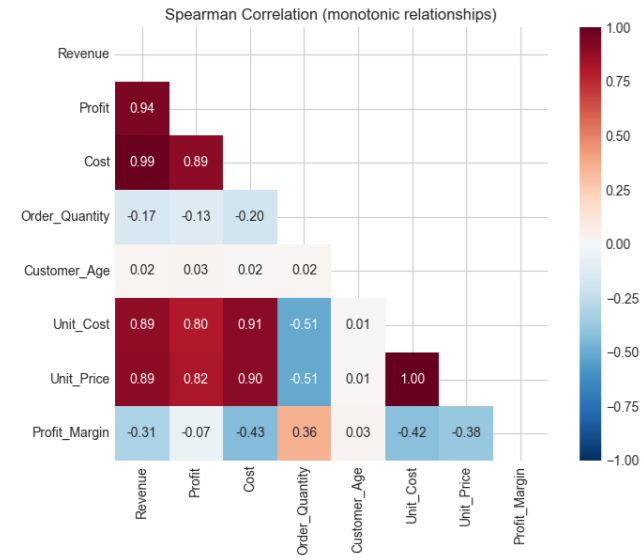
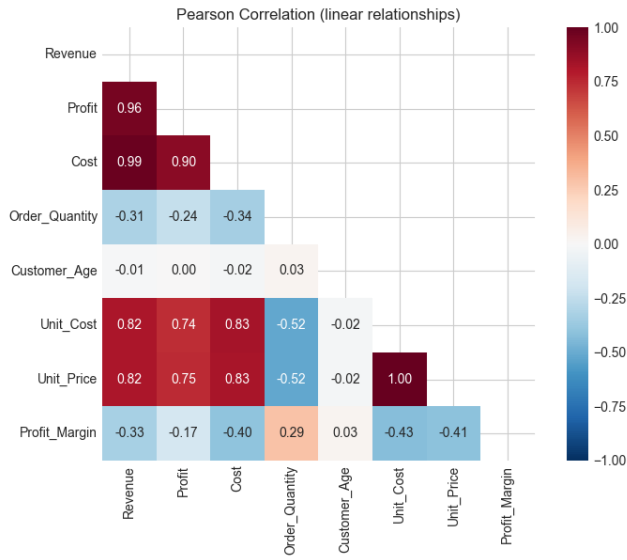
Time series



Distribution Analysis

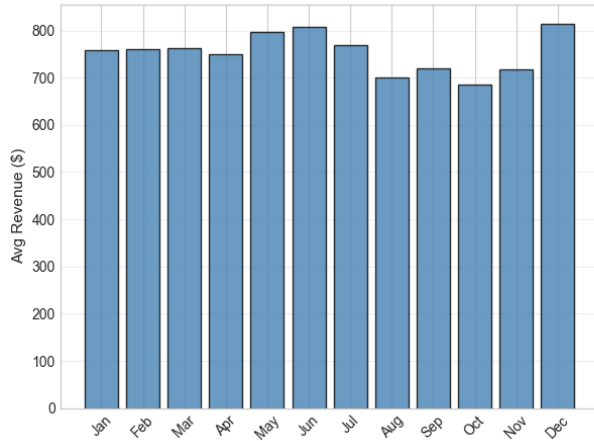


Correlation Analysis

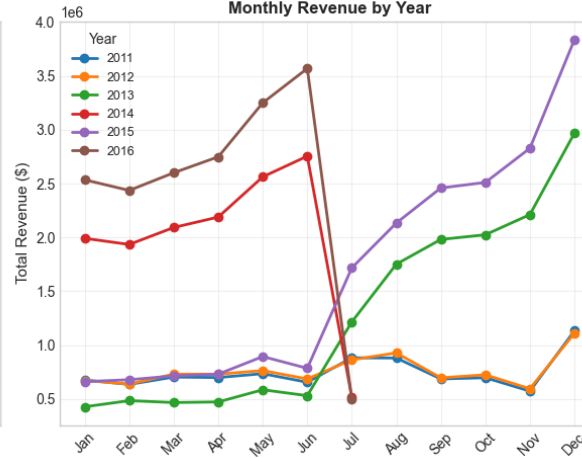


Monthly Pattern Analysis

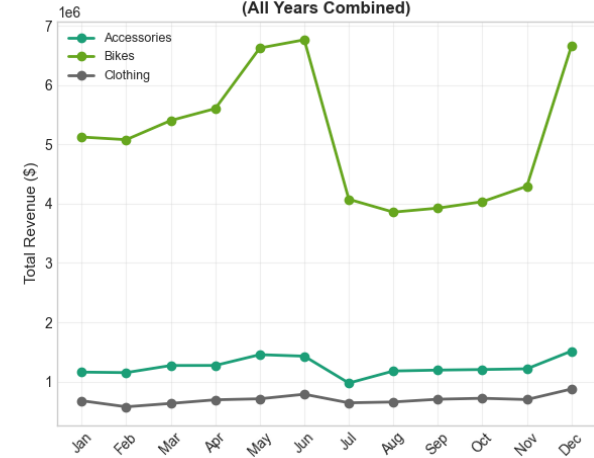
Average Revenue by Month
(All Years)



Monthly Revenue by Year

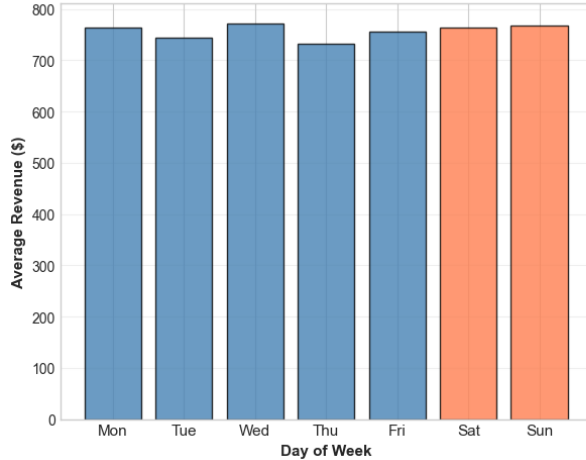


Monthly Revenue by Product Type
(All Years Combined)

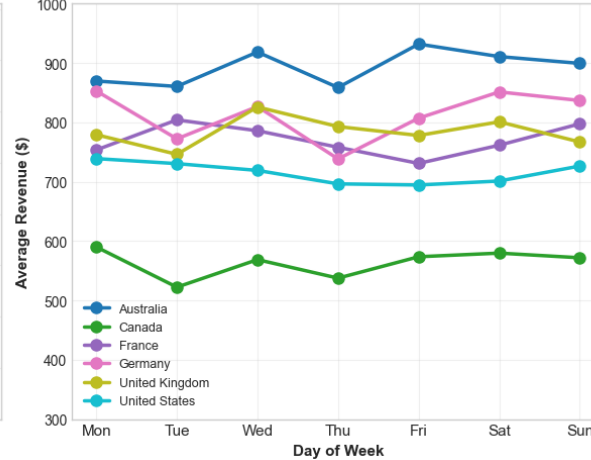


Weekly Pattern Analysis

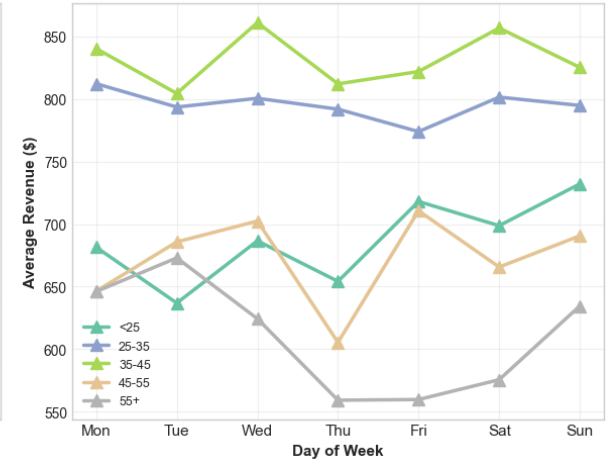
Overall Average Revenue by Weekday



Average Revenue by Weekday and Country

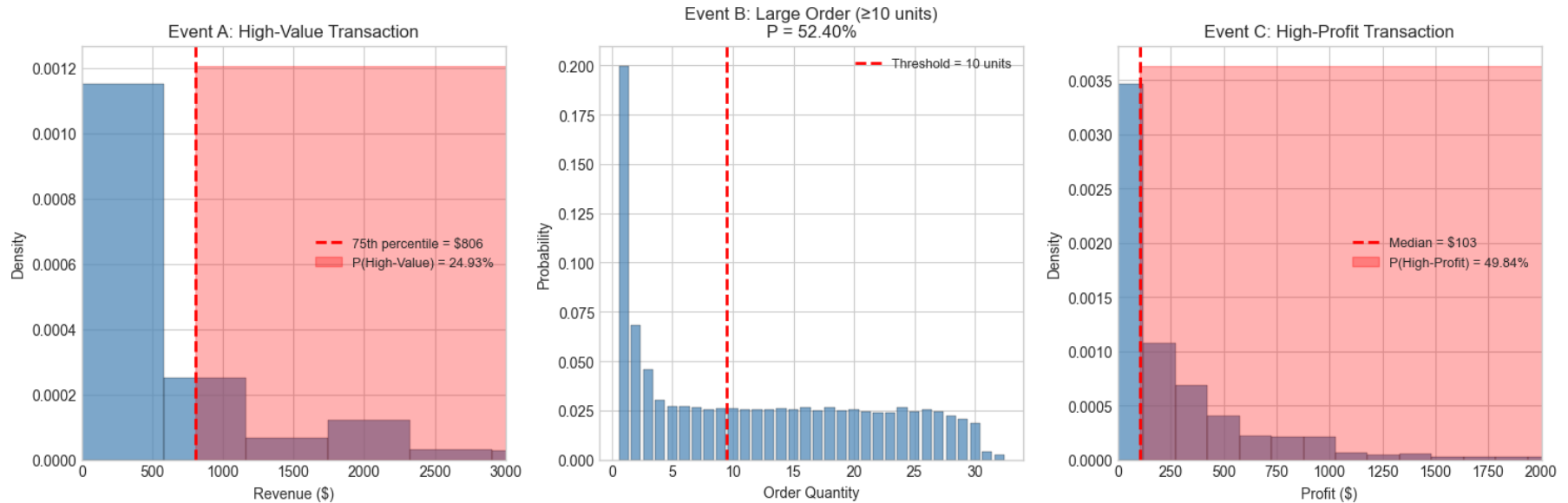


Average Revenue by Weekday and Customer Age

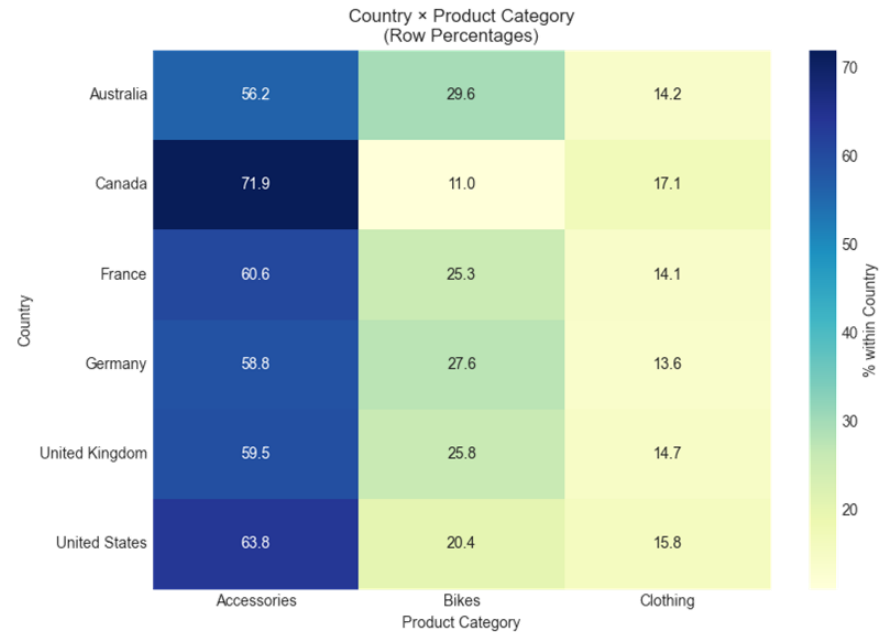
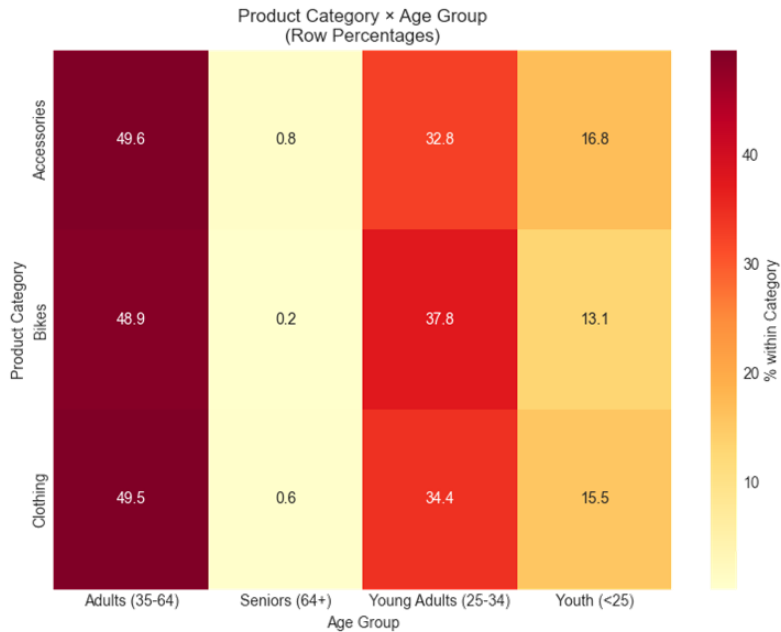


PROBABILITY ANALYSIS

Threshold-based PA



Cross Tabulation Analysis

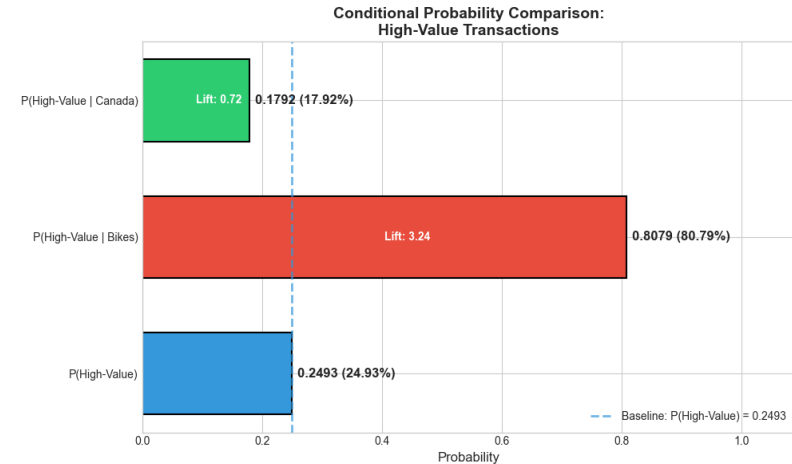


Conditional Probability

$P(\text{High-Value}) = 24,93\%$

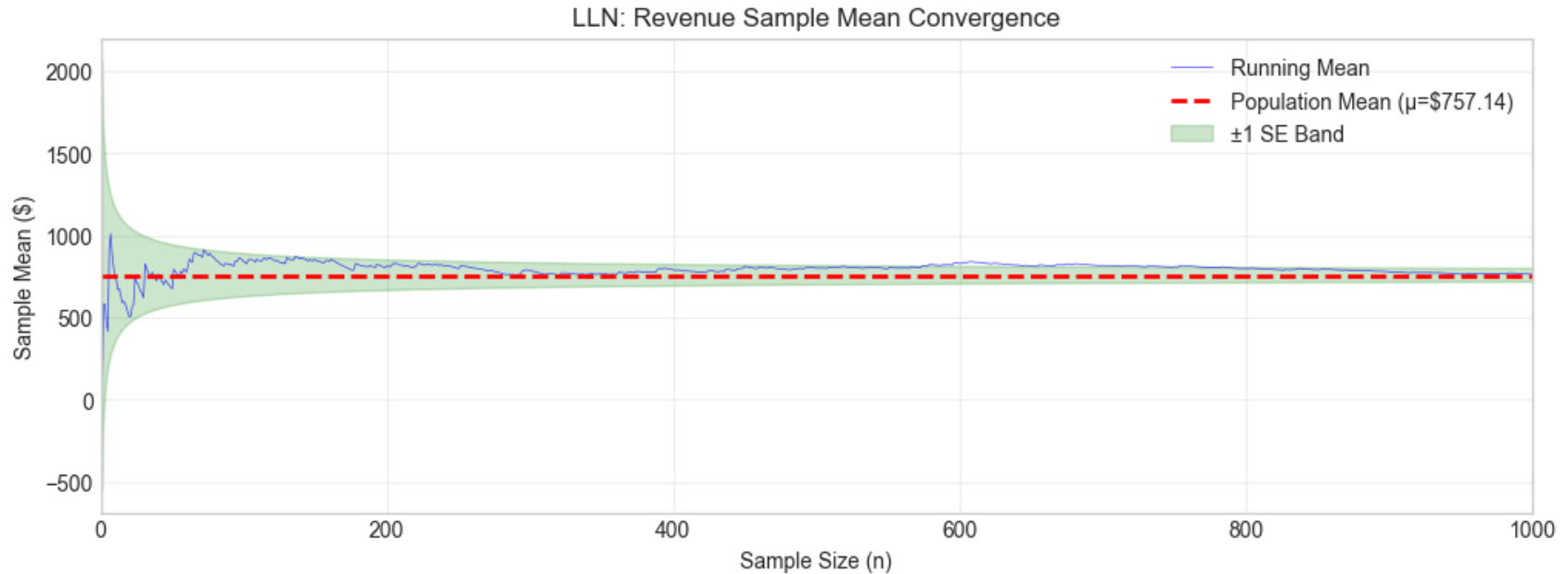
$P(\text{High-Value} \mid \text{Bikes}) = 80,79\%$

$P(\text{High-Value} \mid \text{Canada}) = 17,92\%$



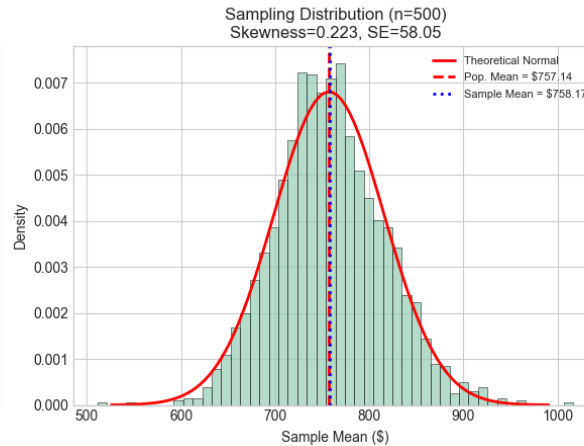
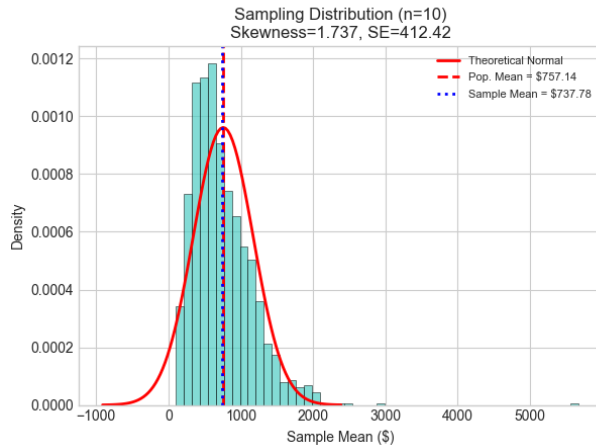
STATISTICAL THEORY APPLICATIONS

Law of Large Numbers

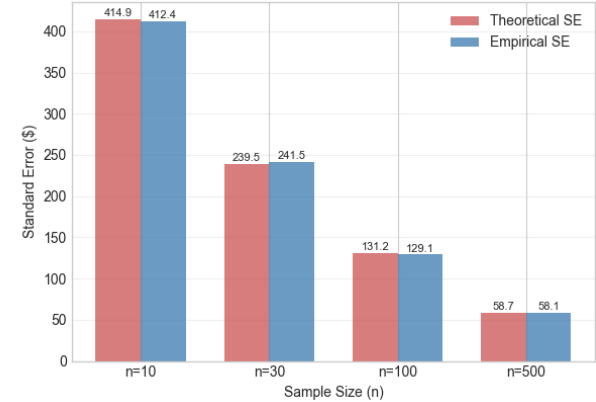


Central Limit Theorem

CLT: Revenue Sample Mean Distribution



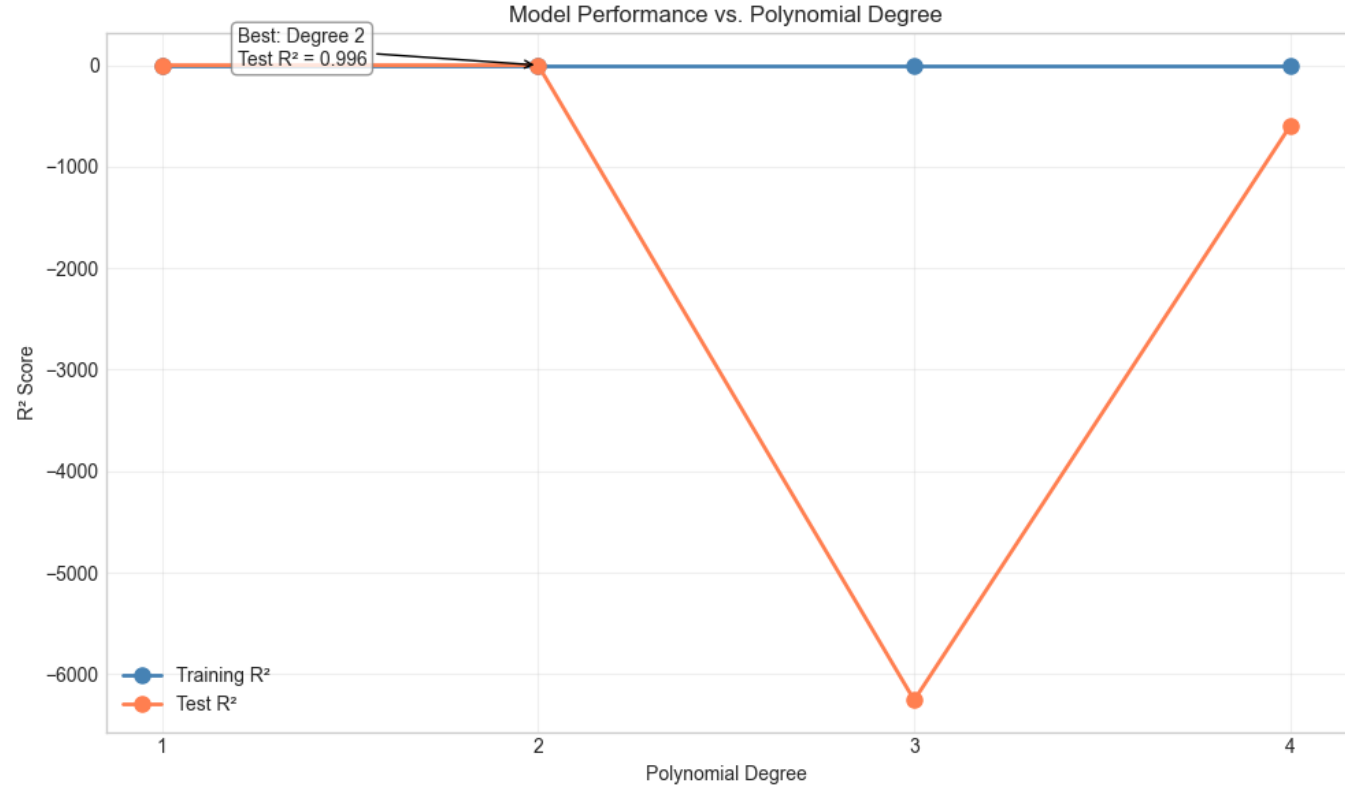
CLT: Standard Error Decay



REGRESSION ANALYSIS

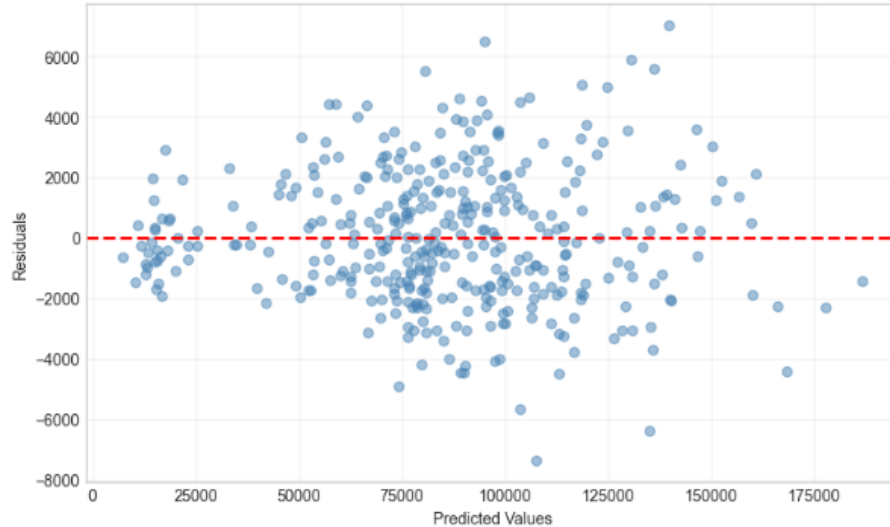
Model Selection

- Compared polynomial degrees 1–4
- Degree 2 achieves best test $R^2 = 0.996$

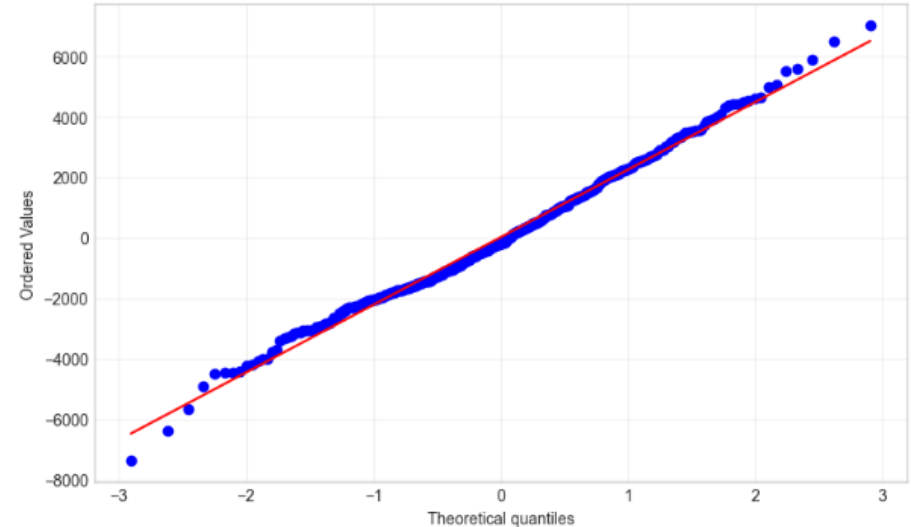


Model Validation

Residuals vs Predicted (Test Set)

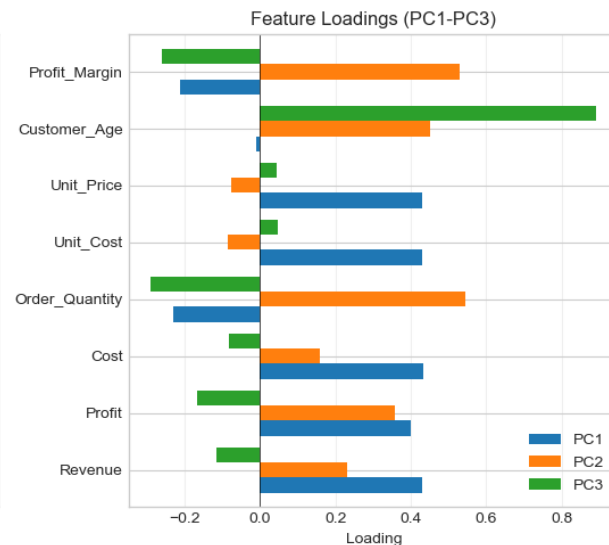
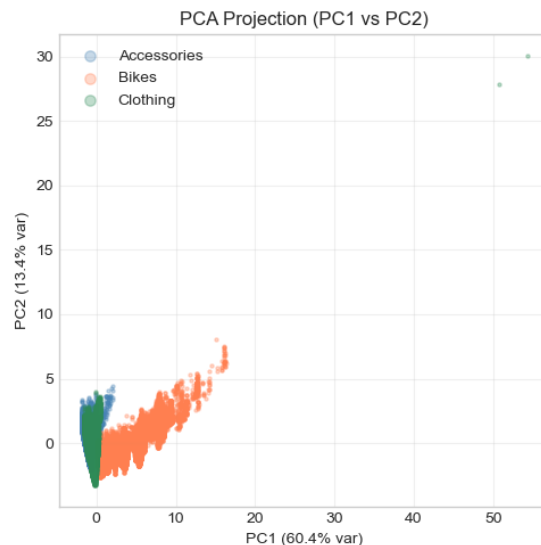
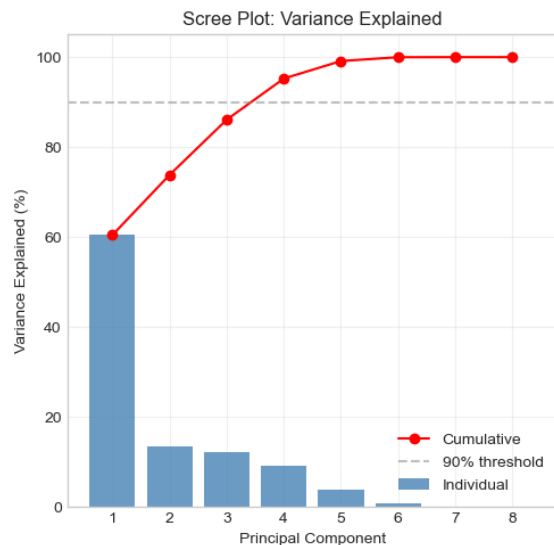


Q-Q Plot (Normality)



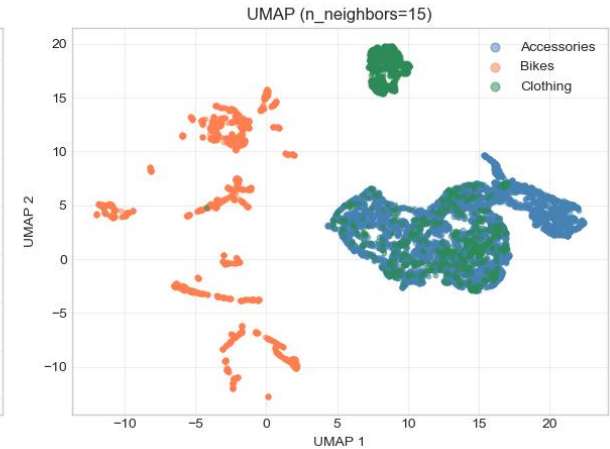
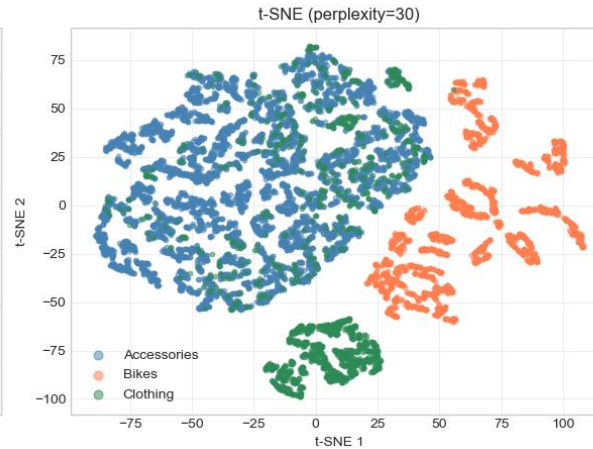
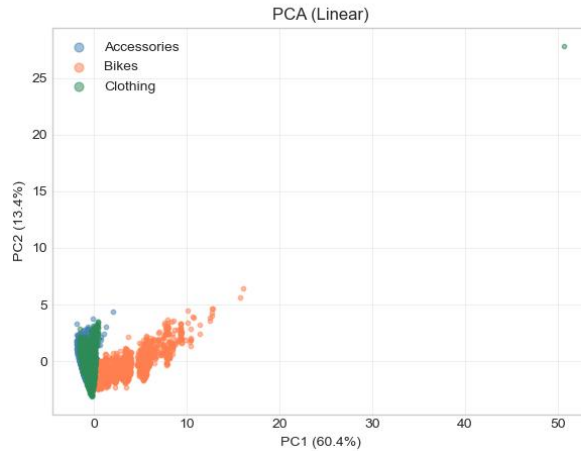
DIMENSIONALITY REDUCTION

PCA



PCA, t-SNE and UMAP

Dimensionality Reduction Comparison: PCA vs t-SNE vs UMAP



Q&A